Introduction to Backtracking Pattern

Backtracking

Backtracking is an algorithmic technique that uses **brute-force** approach to solve a problem.

> Brute-force approach states that for any problem, we should try out all possible solutions and pick up those solutions that satisfy the problem constraints.

In backtracking, we build a solution incrementally and follow the approach that if the current solution can't lead to a valid solution, abandon it and backtrack (or go back) to try another solution. Because of this, recursion becomes a suitable technique for solving backtracking problems.

Dynamic Programming (DP) uses a similar approach where we try out all possible solutions (using Memoization) to pick up the most optimized solution. DP is used to solve optimization problems; backtracking, on the other hand, is mostly used when multiple valid solutions are possible for a problem.

An Example

Let's understand backtracking with an example.

Imagine we want to plant, in a straight line, three fruit trees Apple, Orange, and Mango. What are the possible ways these trees can be planted? The only restriction we have is that Apple and Orange trees can't be placed next to each other.

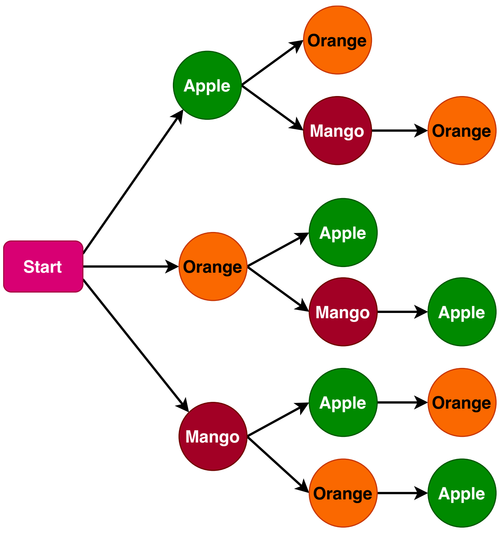
Following a brute-force approach, we can find all possible combinations in which the three trees can be placed and pick those that fulfill the given constraint.

Let's try finding all the possible arrangements.

The first tree can be of any kind, Apple, Orange, or Mango. If our first tree is Apple, the second tree can be Orange or Mango. If we place Orange as our second tree, then our constraint is not fulfilled (which was not to place Apple and Orange together). Hence it is not a valid arrangement; therefore, we go back (i.e., backtrack) and place Mango as our second tree.

Now, we can place Orange in the third place; this arrangement fulfills the problem constraint.

The following diagram shows all possible arrangements:



Backtracking example

From the above diagram, we can clearly see that the two valid arrangements are: `[Apple, Mango, Orange]` and `[Orange, Mango, Apple]`.

This approach of evaluating all possible solutions, going back whenever we see that a certain constraint can't be met, and trying out other possible solutions is called backtracking.

Let's solve some more problems to develop a deeper understanding of backtracking.

← Previous
Solution: Problem Challenge 2: Target Sum

Next →
Combination Sum (Medium)

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